

JAKE FRISCHMANN

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EDUCATION

University of Maryland, College Park

Expected May 2028

B.S. in Computer Science and B.S. in Physics (Dual Degree) **GPA: 3.85, Dean's List 2025**
Minor in Quantum Science and Engineering; Accelerated B.S./M.S. in Computer Science (2028–2029)

Fantasy Book Club President, AppDev Henson Day Initiative, Bitcamp Quantum Organizer, Robotics Terafromers Software

EXPERIENCE & RESEARCH

Zheng Research Lab, Drexel University *Research Assistant* *Department of Mechanical Engineering and Mechanics*

May 2025 – Jul 2025 Philadelphia, PA

- Designed soft robotic manipulators (SpiRob) in SolidWorks to grasp complex geometries via tendon-driven continuum structures and tentacle-style grippers, and built GPU-accelerated simulations of actuated cables and soft-body dynamics in NVIDIA Isaac Gym. Improved grasping non-standard objects compared to current industrial machines.

Pennsylvania Governor's School for the Sciences (PGSS) *Researcher*

Summer 2024 Carnegie Mellon University

- Led a BlokusDuo deep RL team, building PPO + MCTS self-play agents with CNN board encodings that achieved ~85% win rate over tuned heuristics across 18,000+ GPU games and co-authored a 20-page peer-reviewed PGSS Journal paper on the architecture and experiments.

Directed Anxious Generation Correlational Study *Lead Researcher*

Aug 2024 – Dec 2025

- Led a stratified, cluster-sampled survey of high school students ($n \approx 140$) linking multi-platform screen time metrics to validated measures of social connectedness and sleep, cleaning heterogeneous iOS/Android usage data and performing linear regression/t-tests ($\alpha = 0.01$) that found statistically significant negative associations between weekly phone use, sociability, anxiety, and weekday sleep.

Drexel Bioengineering Summer Program *Research Intern*

Summer 2022 Philadelphia, PA

- Prototyped AR-enhanced physical-therapy goggles in Blender, exploring motion-tracking and feedback.

PROJECTS

HensonDay — AR Campus Events App

ARCore, SwiftUI, iOS

- Building an augmented reality mobile app for UMD's Henson Day with the AppDev club. Players discover campus events through location-aware AR creatures tied to real physical spots on campus. ARCore handles world tracking and spatial anchoring; SwiftUI drives the native interface. Production-level club engineering.

Face Tracking Space Invaders — 1st Place, Hackstoga 2023

Python, OpenCV, dlib, cvzone

- Webcam-based dodge game with no keyboard and no mouse: your body is the controller. Background subtraction turns your silhouette into a live collision mask. A dlib correlation tracker locks onto your face and overlays a spaceship. Smiling when hit gives a 50% chance to survive.

Orbit UMD

SvelteKit, TypeScript, Supabase, GitHub Actions

- Built an academic planning platform for UMD students with course lookup, degree-audit flows, four-year planning tools, and profile-aware guidance.

RoadTrip: AI-Enhanced Vacation Planning iOS App

SwiftUI, GPT-4, REST APIs

- Shipped a SwiftUI iOS app that orchestrates GPT-4, Google Maps, and Booking.com APIs to generate multiday itineraries, with CoreLocation/CoreData-backed offline storage and resilient networking (retries, rate-limit handling, caching) for smooth performance on unreliable networks.

TOOL: Music Generation & Style Transfer Platform

Python, PyTorch, TensorFlow

- Engineered a TOOL-style music generation platform by scraping audio/lyrics metadata and training U-Net-style generative models, using mixed-precision training and TensorBoard-tracked hyperparameter sweeps to iterate on style-transfer architectures.

Medication Management and Adherence App

React Native, TypeScript, HCL

- Designing a cross-platform tracker to increase medication adherence by applying friction-in-design, habit formation, and progressive disclosure to lower cognitive load and surface adherence patterns while preserving user privacy.

TECHNICAL SKILLS

Languages Python, C++, C#, Rust, Swift, JavaScript, TypeScript, SQL, R

Project Management GitHub, GitLab, GitHub Actions, Jira, pytest, Prettier, CMake

Machine Learning Coursera Deep Learning Specialization, PyTorch, TensorFlow, Hugging Face Hub, Kaggle

Computer Vision OpenCV, dlib, Haar Cascades, ARCore, Background Subtraction

Web and Cloud MERN stack, React Native, SvelteKit, REST APIs, OAuth, .NET MAUI, SwiftUI, Supabase

Robotics SolidWorks, NVIDIA Isaac Gym, G-code, Soft Robotics Simulation

REFERENCES

Available upon request, including:

- Dr. Yue (Luna) Zheng, Mechanical Engineering Professor, Drexel University yz894@drexel.edu
- Mollie Kaufer, Communications Professor, University of Maryland mkaufer@umd.edu